EMS Incident Dispatch

Data Warehouse and Analytics

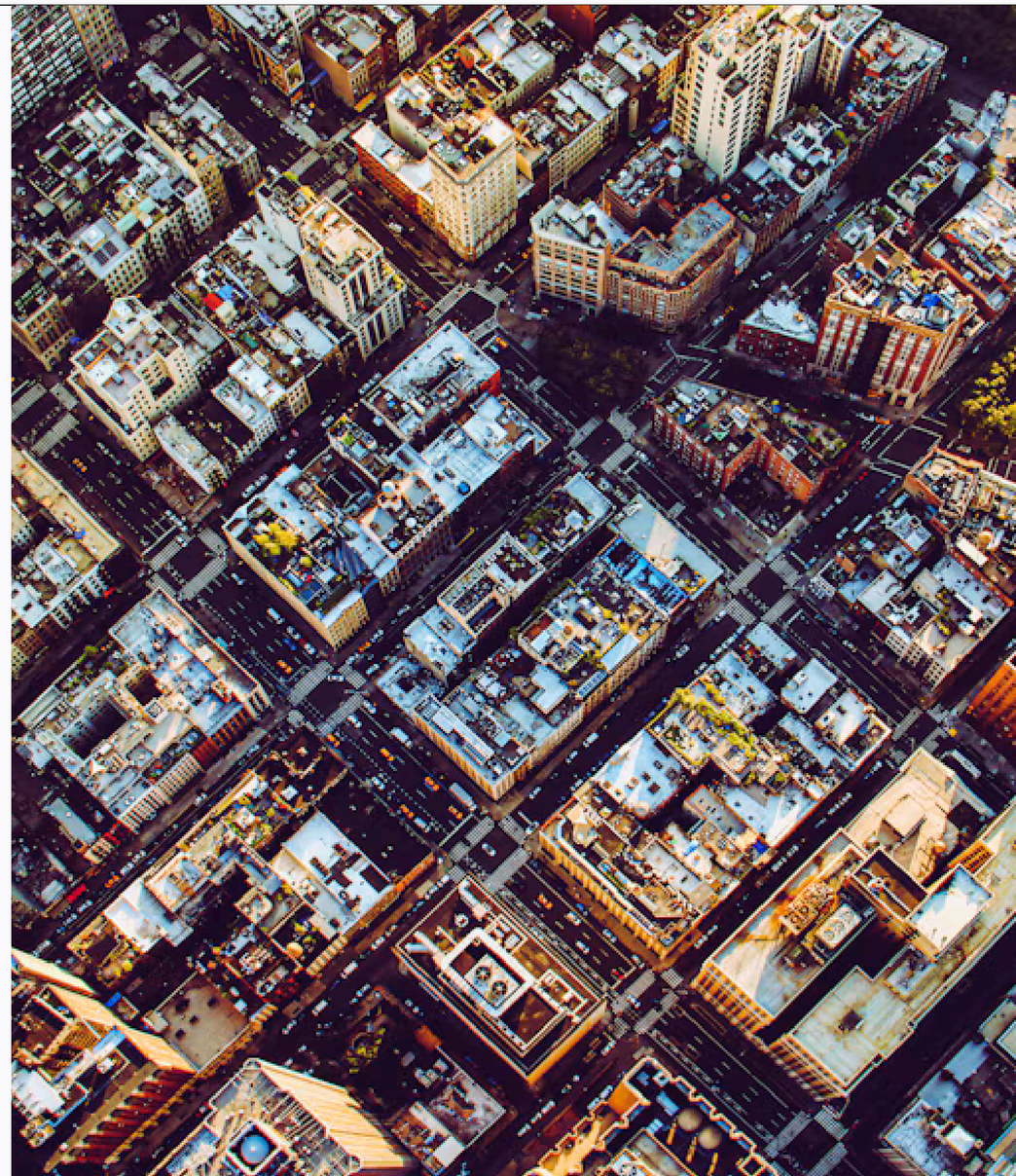


A data investigation performed for The Committee on Fire and Emergency Management, New York City Council

The Committee on Fire and Emergency Management, New York City Council can fund a limited number of new ambulance tours this year.

Members want to put them where they will do the most good for people in life-threatening emergencies.

They want to perform an inquiry to identify where to place these new tours.



Report Objectives

The committee wants to know what the data says before choosing where to allocate new ambulance tours.

01 How have response times for the most serious calls changed year by year?

02 Which boroughs and dispatch areas are slowest for the most serious calls?

03 Is the delay in dispatching (call to ambulance assigned) or in travel (assigned to on scene)?

04 When are responses slowest, by hour of day and day of week?

05 Did areas that received extra ambulance capacity improve afterwards?

EMS Incident Dispatch Dataset

Provided by Fire Department of New York City



Total rows

30.4M



Columns:

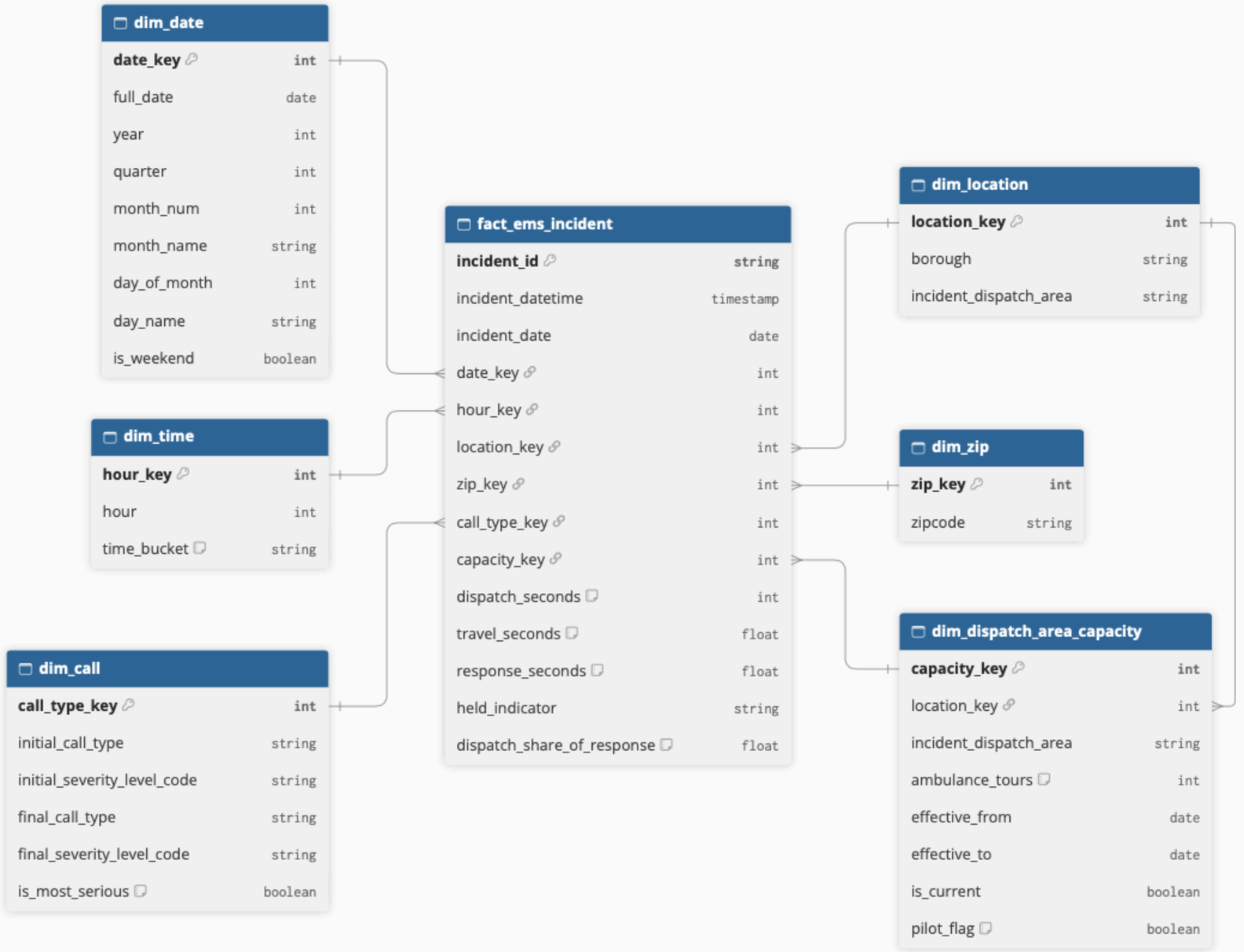
31

Key Fields

- INCIDENT_DATETIME
- BOROUGH
- INCIDENT_DISPATCH_AREA
- SEVERITY_LEVEL_CODE
- DISPATCH_RESPONSE_SECONDS

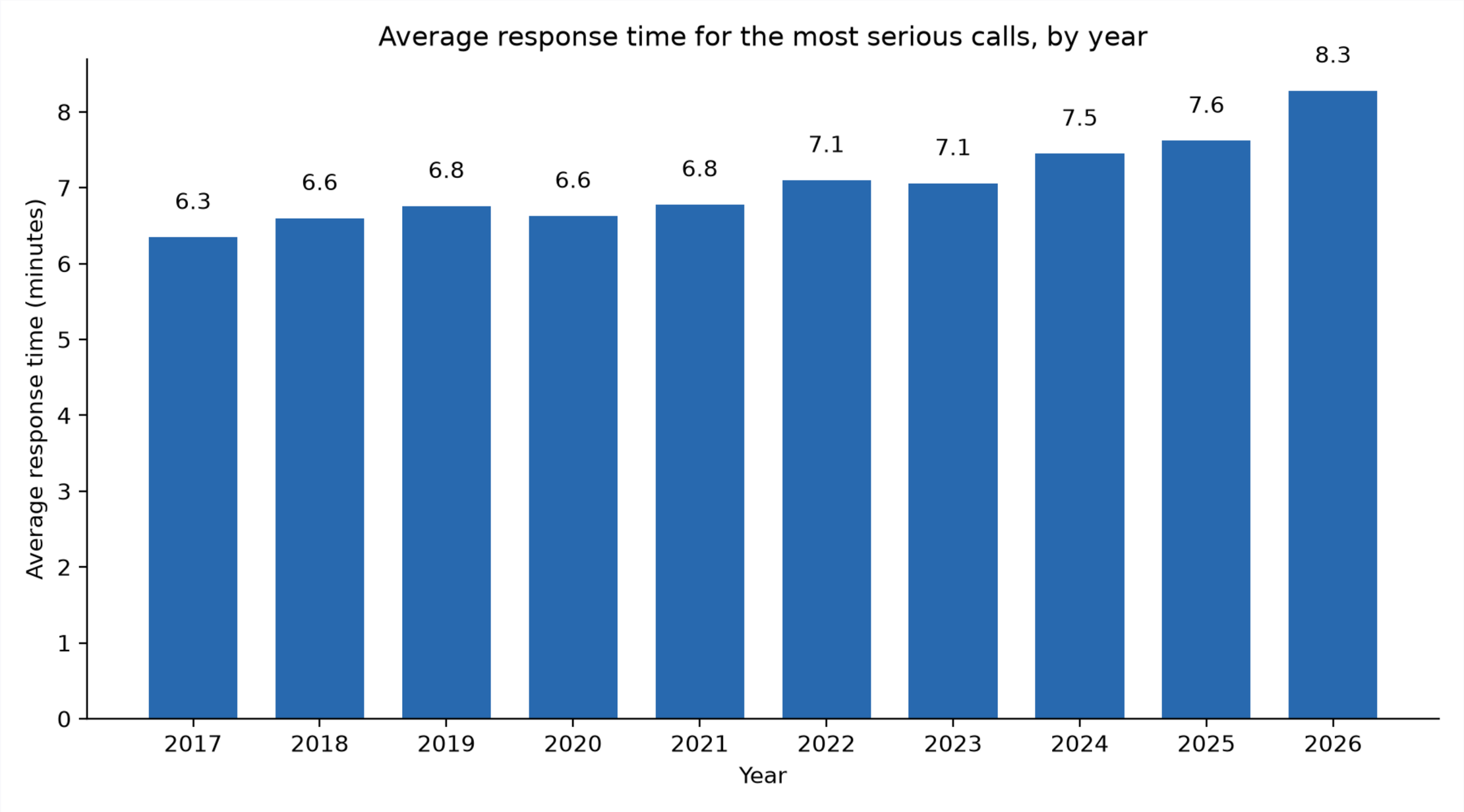
The data covers information about the incident as it relates to the assignment of resources and the Fire Department’s response to the emergency.

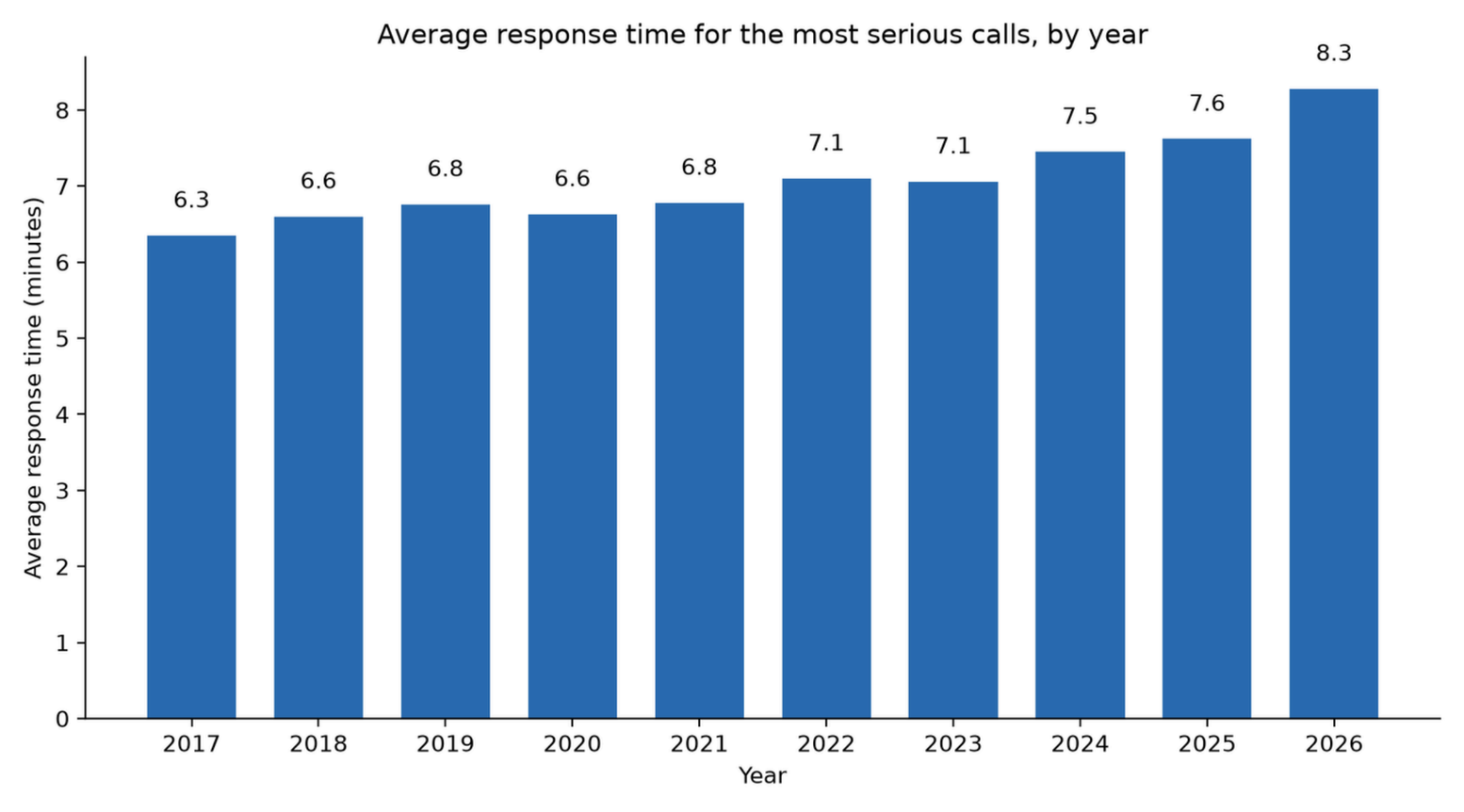
DATA ORGANIZATION



Question 1

How have response times for the most serious calls changed year by year?


Average response time for the most serious calls, by year
Average response time (minutes)
0
1
2
3
4
5
6
7
8
6.3
6.6
6.8
6.6
6.8
7.1
7.1
7.5
7.6
8.3
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Year



CONCLUSION

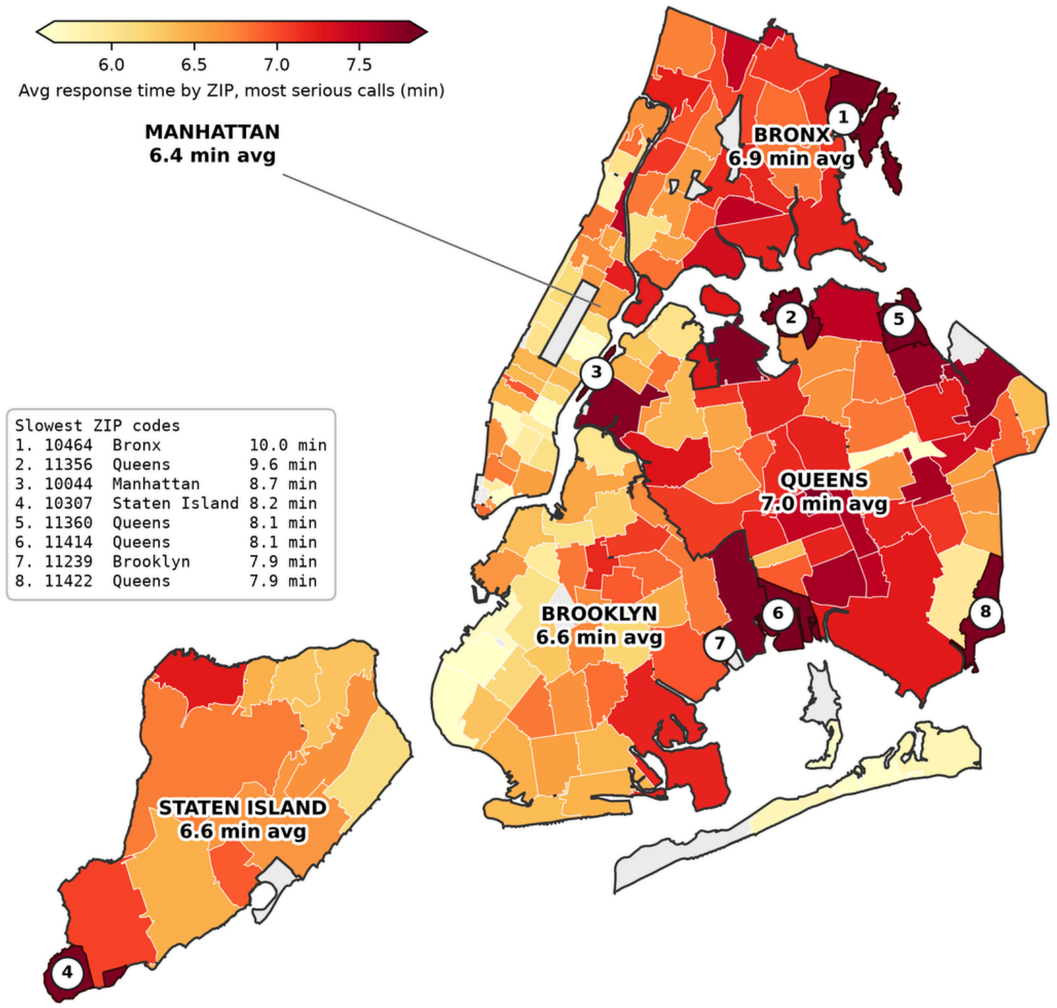
Response times for the most serious calls have risen about 2 minutes since 2017.

Question 2

Which boroughs and dispatch areas are slowest for the most serious calls?

Which Boroughs and Dispatch Areas Are Slowest for the Most Serious Calls?

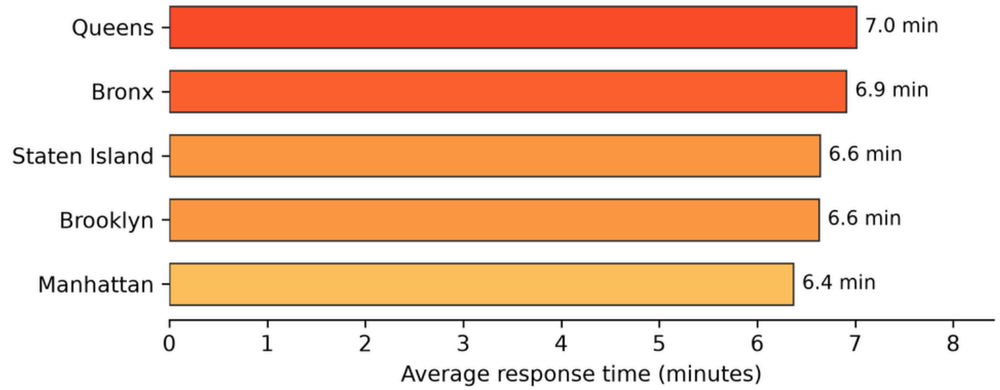
Where Are Response Times Slowest?



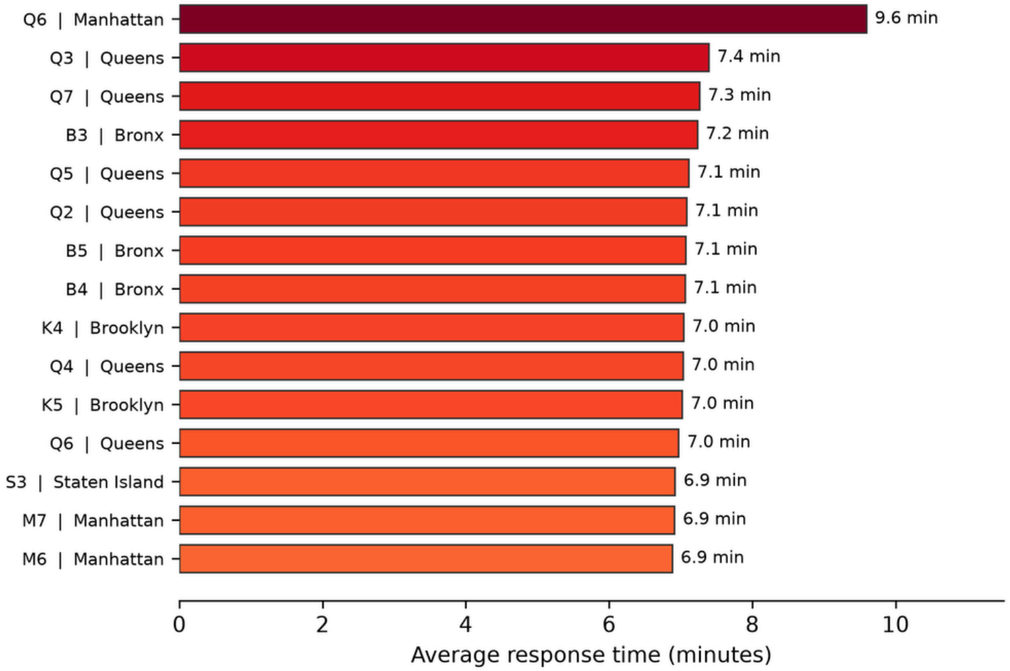
Slowest ZIP codes

1.	10464	Bronx	10.0 min
2.	11356	Queens	9.6 min
3.	10044	Manhattan	8.7 min
4.	10307	Staten Island	8.2 min
5.	11360	Queens	8.1 min
6.	11414	Queens	8.1 min
7.	11239	Brooklyn	7.9 min
8.	11422	Queens	7.9 min

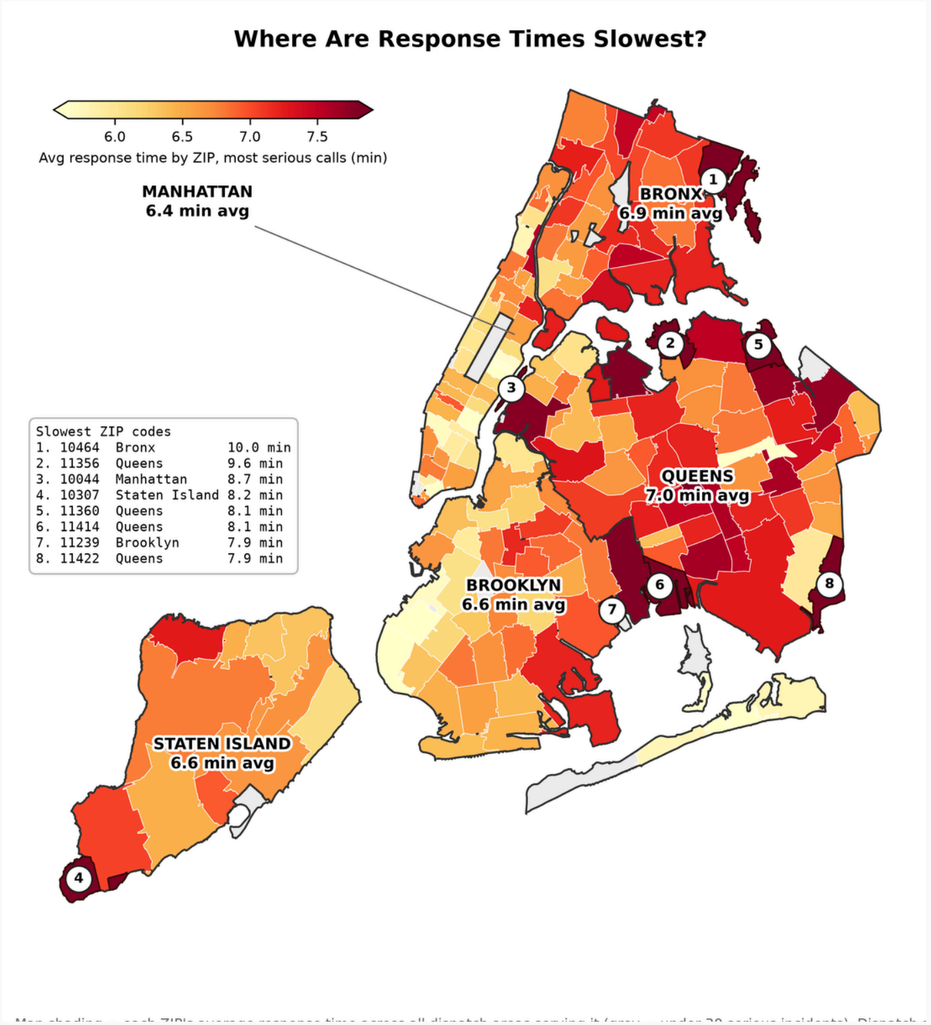
Boroughs: Slowest at Top



15 Slowest Dispatch Areas (citywide)



Map shading = each ZIP's average response time across all dispatch areas serving it (grey = under 30 serious incidents). Dispatch areas have no published boundaries, so they are ranked in the bar chart. Colour scale clipped at the 5th/95th percentile.



CONCLUSION

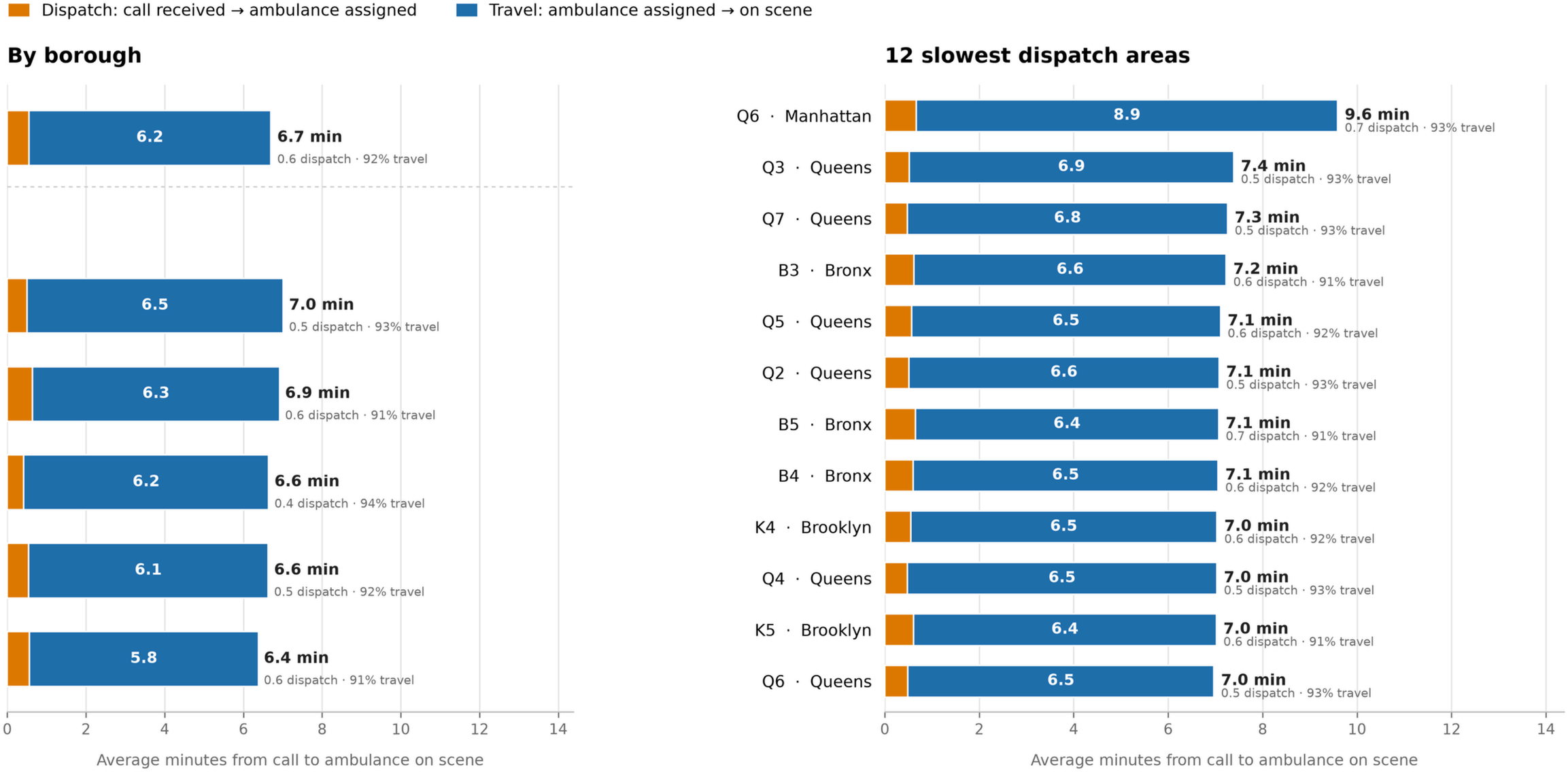
Queens (7.0 min) and the Bronx (6.9 min) are the slowest boroughs, and dispatch area Q6 in Manhattan stands out at 9.6 minutes, well above every other area.

Question 3

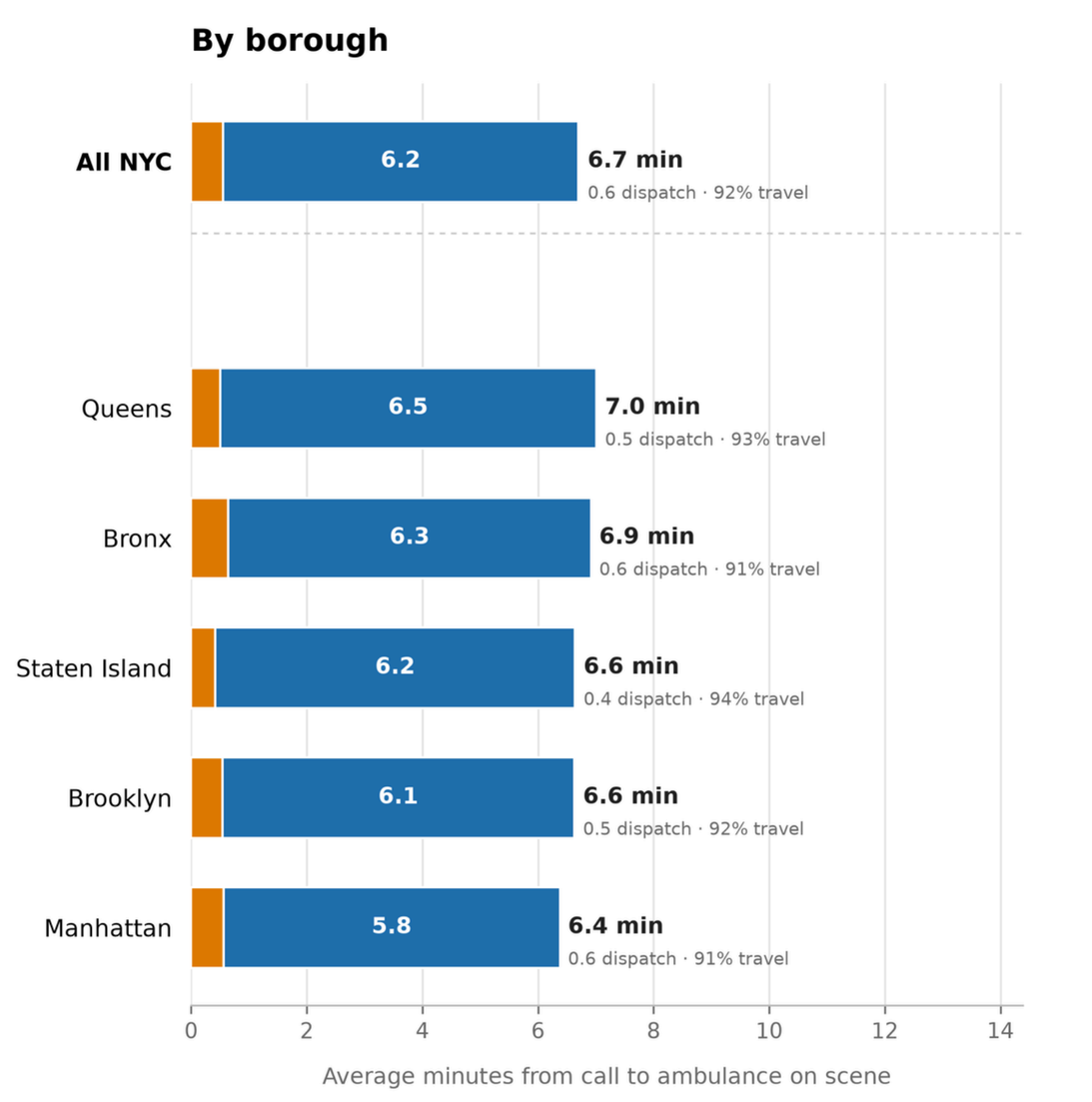
Is the delay in dispatching (call to ambulance assigned) or in travel (assigned to on scene)?

Is the delay in dispatching or in travel?

Travel accounts for 92% of the 6.7-minute average response to the most serious calls. Dispatch takes about 33 seconds.



Most serious calls only (final severity < 3); valid response times only. Dispatch areas with fewer than 30 incidents excluded. Travel includes crew turnout after assignment.

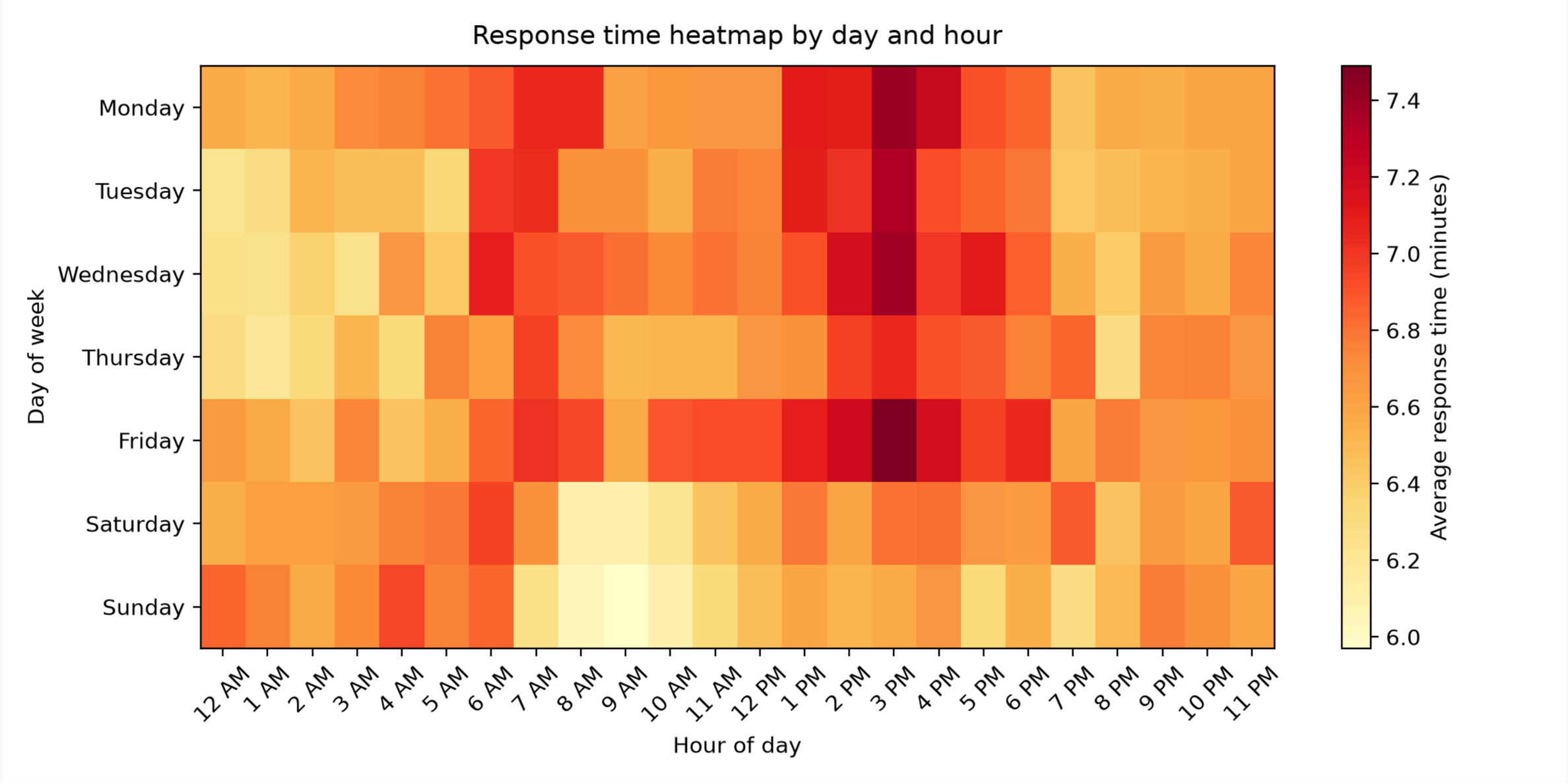


CONCLUSION

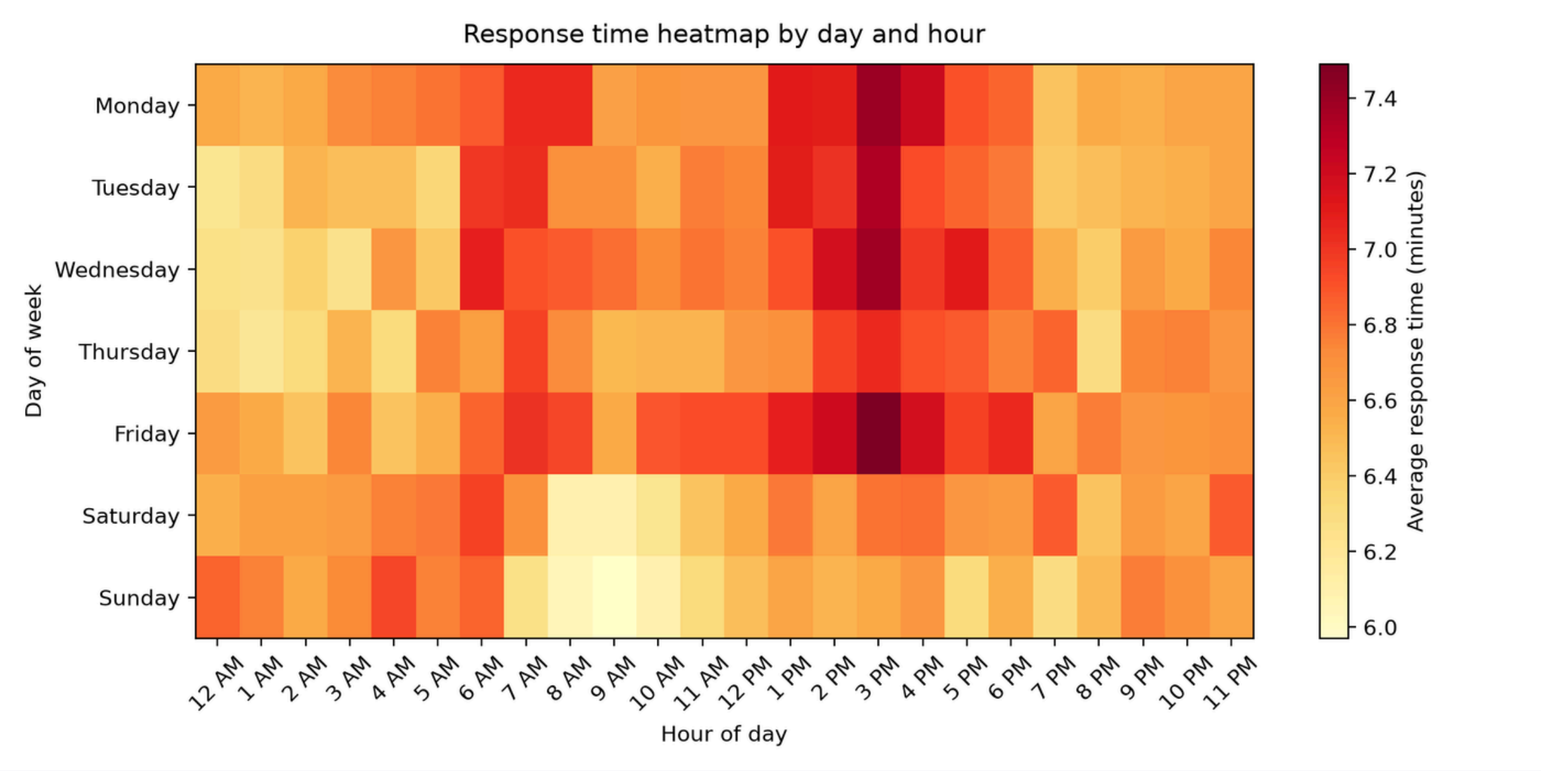
Travel is the delay: about 92% of the average response, versus roughly 33 seconds for dispatch

Question 4

When are responses slowest, by hour of day and day of week?


Response time heatmap by day and hour
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of week
12 AM
1 AM
2 AM
3 AM
4 AM
5 AM
6 AM
7 AM
8 AM
9 AM
10 AM
11 AM
12 PM
1 PM
2 PM
3 PM
4 PM
5 PM
6 PM
7 PM
8 PM
9 PM
10 PM
11 PM
Hour of day
7.4
7.2
7.0
6.8
6.6
6.4
6.2
6.0
Average response time (minutes)

When are responses slowest, by hour of day and day of week?



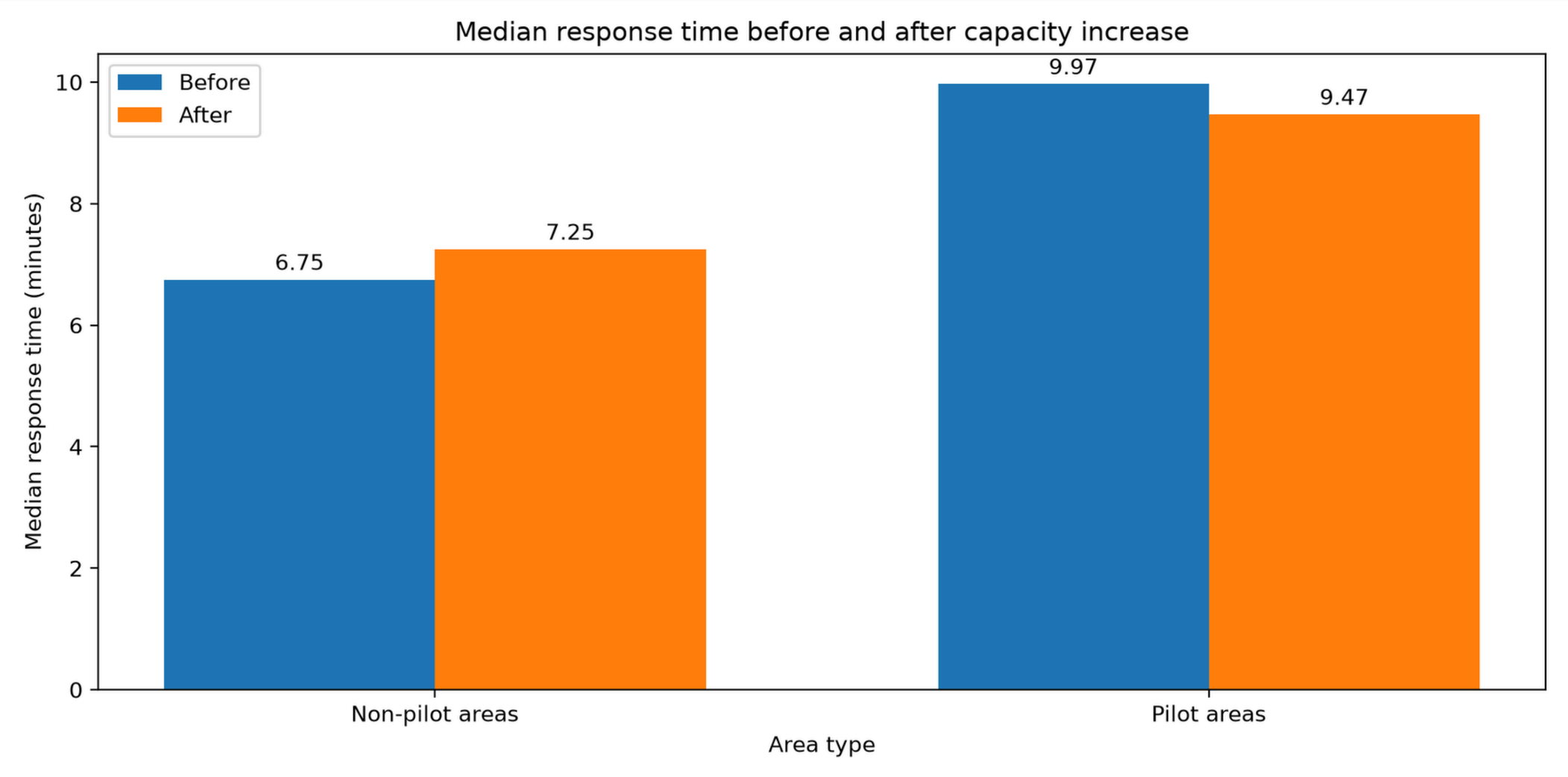
- Weekday afternoons, particularly around 3 PM, show some of the highest average response times.
- Weekend response times generally appear lower.
- Increased weekday traffic and emergency demand may contribute to slower afternoon responses.

Question 5

Did areas that received extra ambulance capacity improve afterwards?

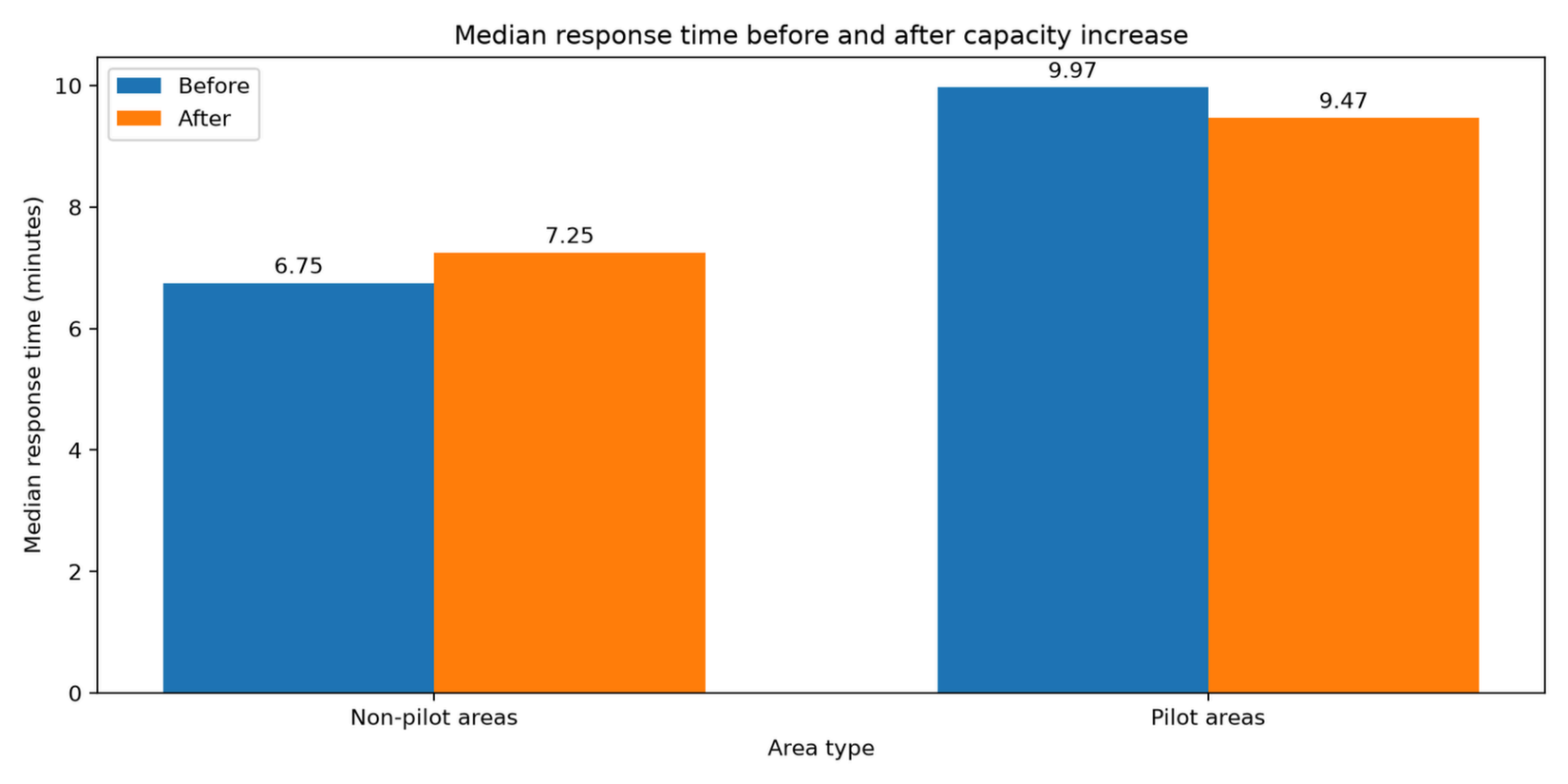
Simulation

- **Goal** - Simulate what would happen if additional ambulance resources were allocated to areas with slower response times.
- The dataset was divided into two periods, using the midpoint as our simulated pilot start date.
- The slowest 20% of dispatch areas by average response time were selected.
- For the selected pilot areas, we simulated a 25% increase in ambulance tours, with at least one additional tour being the minimum.
- This allowed us to compare response times before and after the simulated increase, while also comparing those results against areas that did not receive additional capacity.



Pilot area - Locations where we chose to increase ambulance tours

Did areas that received extra ambulance capacity improve afterwards?



- Pilot areas showed a 0.50-minute decrease in median response time after the simulated capacity increase (9.97 → 9.47 minutes).
- Non-pilot areas experienced a 0.50-minute increase in median response time (6.75 → 7.25 minutes).
- Results are promising but inconclusive due to the smaller pilot sample and the simulated nature of the capacity increase.

Recommendations

- Target travel time, not dispatch.
- Prioritize the slowest areas.
- Add weekday afternoon coverage.
- Put added capacity where response is historically slowest

Thank You.

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